Discovering AI

Raspberry Pi Foundation

Have you heard of artificial intelligence (AI)?



Talk to the person next to you, and describe what you think artificial intelligence is.

Lesson 1: What is AI?

Success criteria

- I can **describe** how modern AI systems make predictions from patterns in data
- I can **identify** uses of AI systems

Artificial intelligence (AI)



Artificial intelligence (AI) is a set of tools used to help solve difficult problems using computers.

AI systems

1. They are created by humans
2. They detect **patterns** in **data**
3. They use **patterns** to make **predictions**

Activity 1

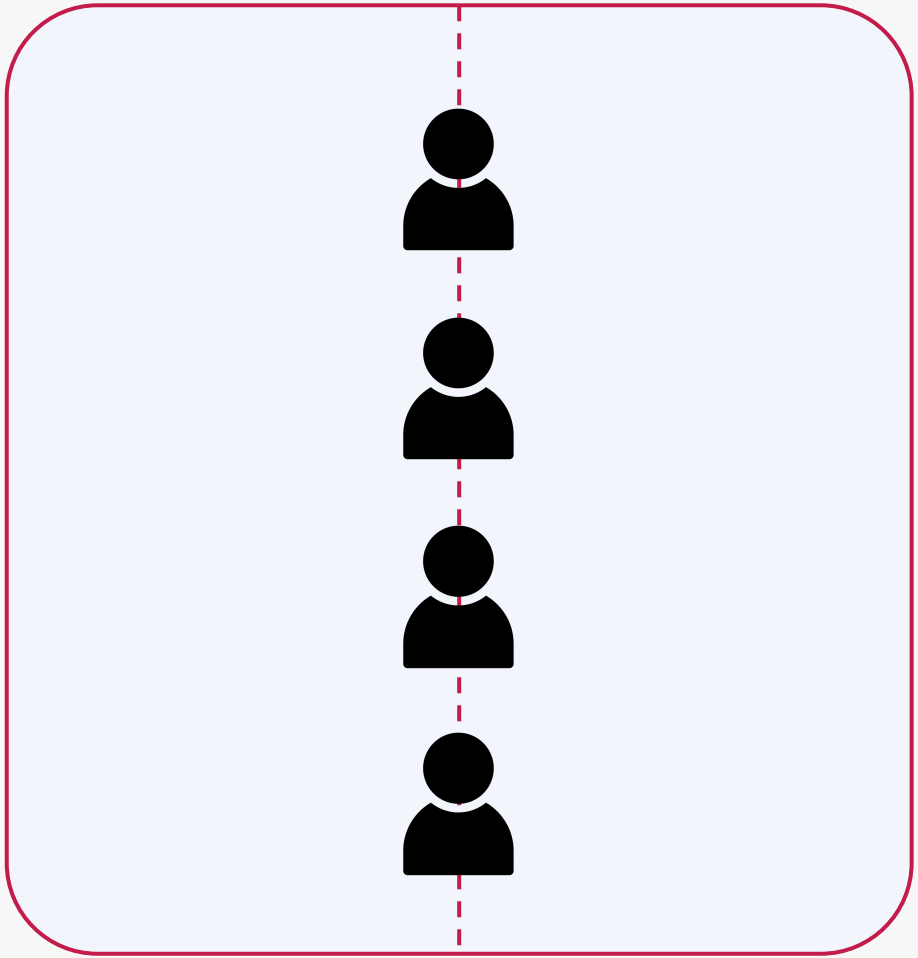
You are going to decide if you think a technology uses AI or not.



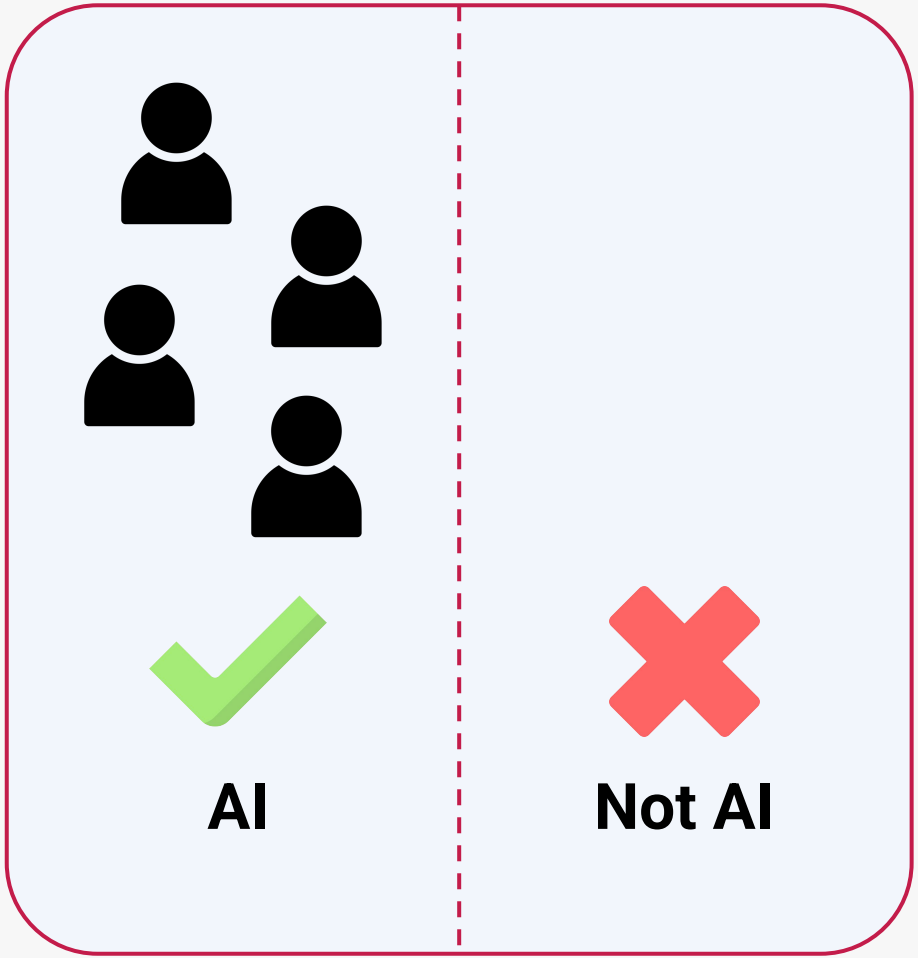




Activity 1

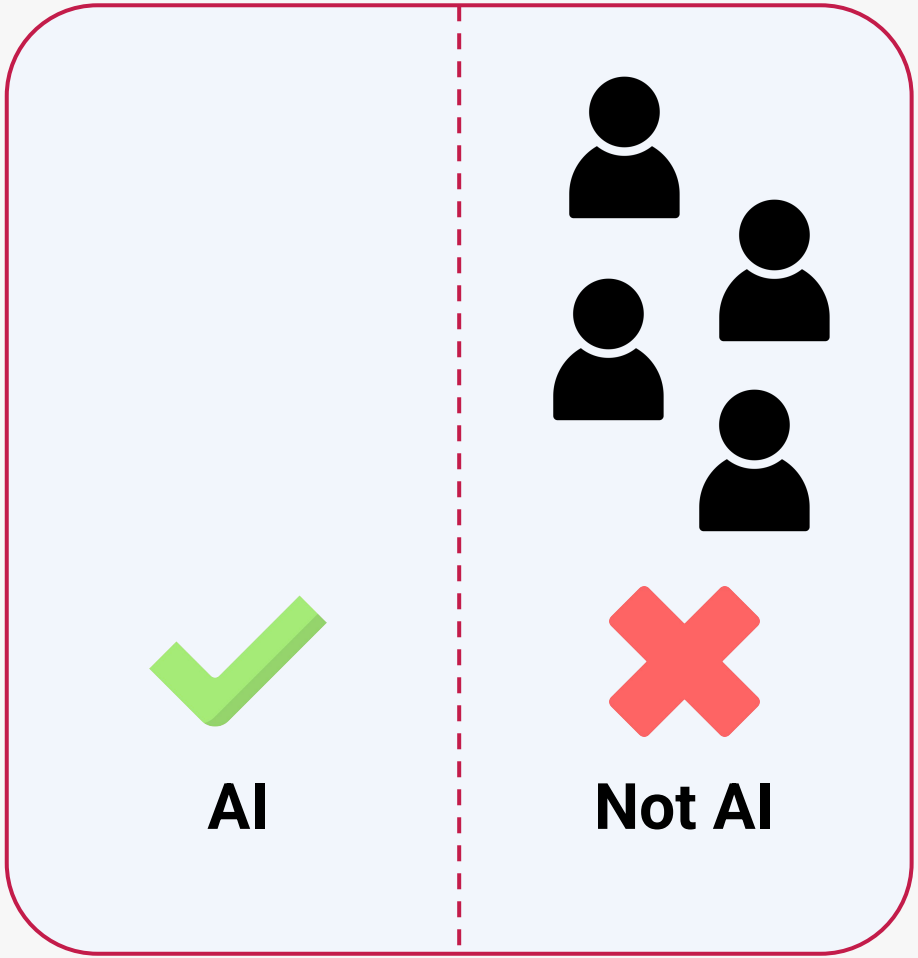


Stand in the centre of the room.



Look at each technology.

If you think it uses AI, move to the AI side of the room.



If you think the technology **does not** use AI, move the other side of the room.

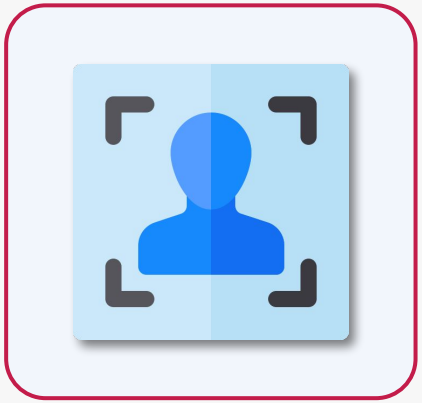
Do these technologies use AI?



Self-driving car



Alarm clock



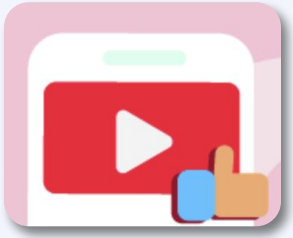
A tool to check who someone is from their face

Uses of AI systems

AI systems can be used to do many **different tasks** using computers.



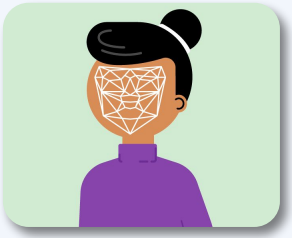
Generating content



Recommending content



Predicting future outcomes



Computer vision

Image generated with AI



Computer games

What is a prediction?



A **prediction** is when you use what you know about the past and present to work out what might happen in the future.

Image generated by AI

Make a prediction



Dragons

0 - 4



Eagles



Eagles

0 - 1



Tigers

Who do you predict will win?



vs



Think – Pair – Share

Your prediction

Your **prediction** was based on **data** about:



Was your prediction correct?

The results are in…



2 - 1



Was your prediction correct?

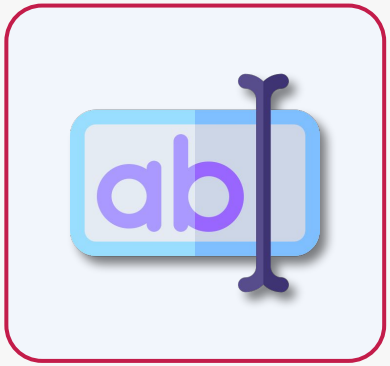
Why might a prediction sometimes be wrong?



Think – Pair – Share

What is data?

Examples of **data** include text, images, numbers, and sound.



Text



Images



Numbers



Sound

Your prediction

Which type of **data** did you use to make your prediction?



Text



Images

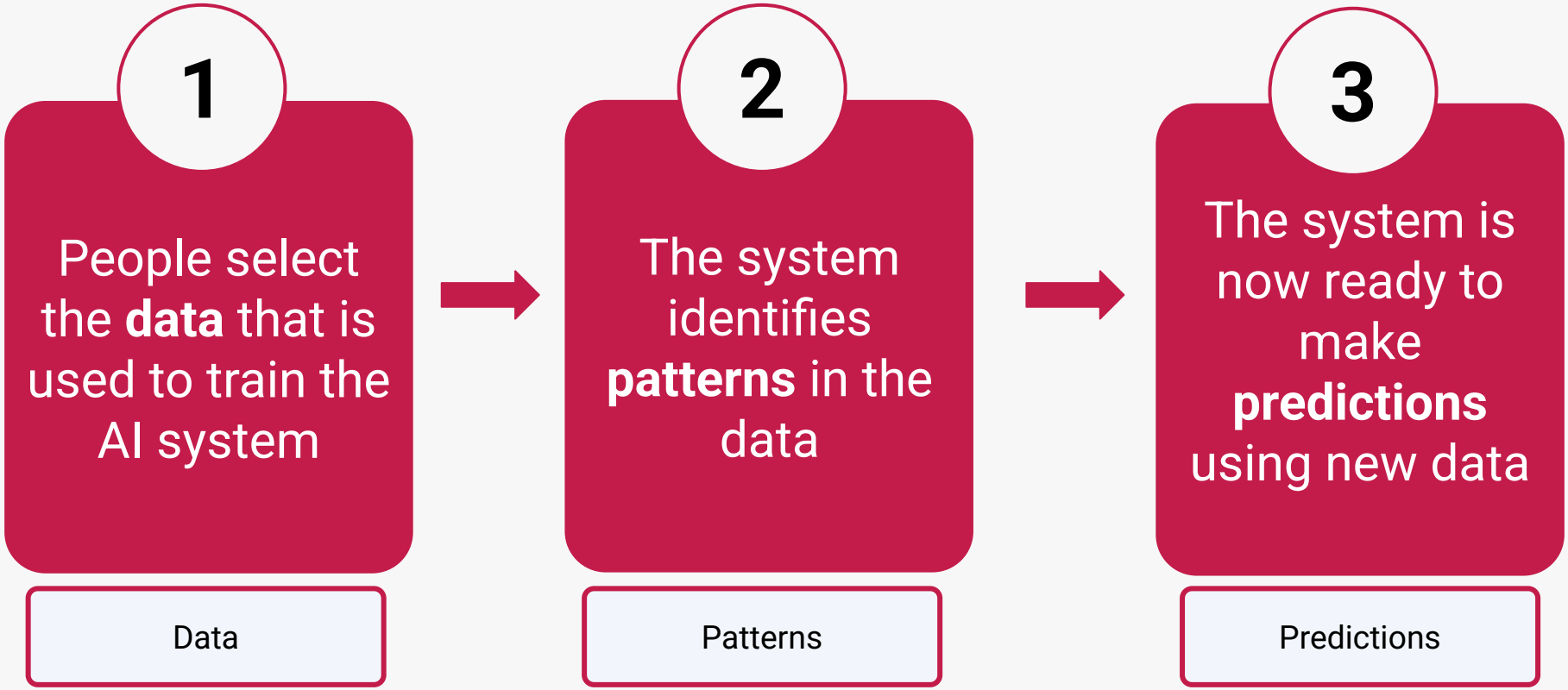


Numbers



Sound

Creating an AI system



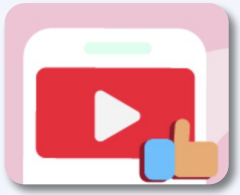
The prediction pathway – Your turn

On your worksheet, draw a line between:

- The data needed to train the AI system
- The AI system
- The prediction made



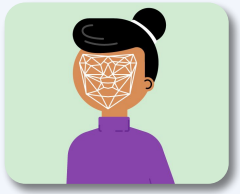
Generating content



Recommending content



Predicting future outcomes



Computer vision

Worksheet answers

Data	AI system	Prediction
Text data, such as online chats, websites, and online books	 Chatbot	The next word or phrase to respond to the question entered by a human

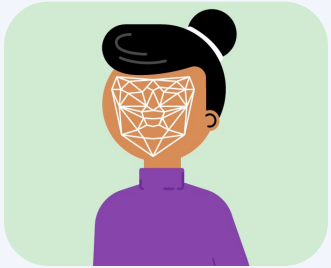
Worksheet answers

Data	AI system	Prediction
What you and people with the same interests as you have liked and watched before	Video recommendations 	Videos you would like to watch

Worksheet answers

Data	AI system	Prediction
Weather data such as temperature and rainfall	 Flood predictions	If an area will flood and how high the water will rise

Worksheet answers

Data	AI system	Prediction
Lots of images of a person's face	 Unlocking a phone with your face	Whose face it is — the phone's owner or someone else

Optional

The prediction pathway: Extension

A satellite navigation system that identifies the fastest route from one place to another



Questions:

1. What **data** does this AI system use to make **predictions**?

2. What **prediction** does this AI system make?

Extension answers

Data	AI system	Prediction
Real-time traffic data, past traffic data, maps	 Satellite navigation	The best way to get from one place to another

What is artificial intelligence (AI)?

AI systems **are not**...

- Always correct
- Robots that think and control themselves
- Something that will replace humans

AI systems **are**...

- Tools and techniques used to help solve difficult problems
- Created by humans
- Tools used by people to help with parts of their job

Exit ticket

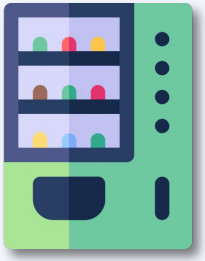
AI or not AI?

Do you think these systems use AI?

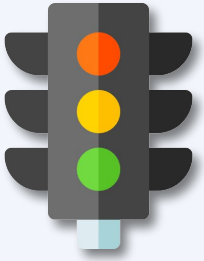
Pick at least one and describe why you think it does or does not use AI.



Voice recognition



Vending machine



Traffic light